Real time neural-fuzzy controller

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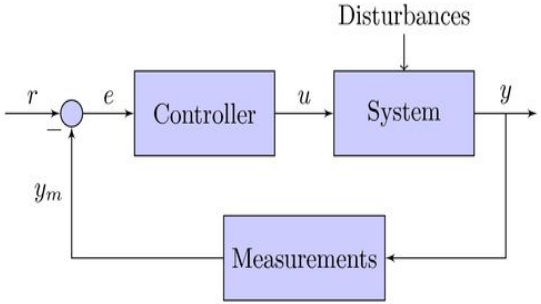
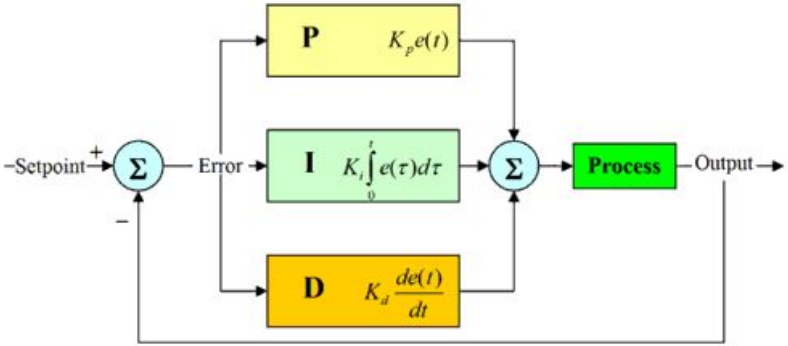
Inspiration

- Control systems are used everywhere
- Most control systems use classical control theory
  - Very difficult to model every system with equations
  - Real control problems are nonlinear
  - Tuning requires experts with many years of experience





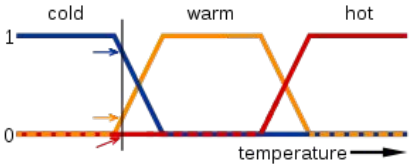
Aim of this IP

- Understand and investigate state of art of neural fuzzy system
- Investigate implementation of neuro-fuzzy system on microcontroller
- Demonstrate a real time control system that uses neuro fuzzy logic on a 1 DOF ball balancer.

Fuzzy Logic

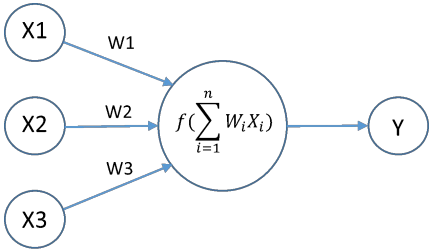
- Based on a system of non-digital set theory and rules. Developed by Lofti Zadeh in 1965.
- It's advantages is its ability to deal with vague systems and its use of linguistic variables and membership functions
- Assigns degree of truth to an input

The temperature is 10% warm and 80% cold



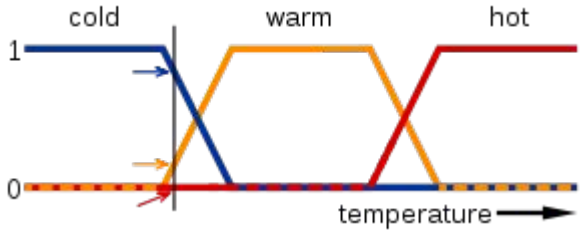
Neural Networks

- Inspired around how the brain works with neurons
- Universal approximators
- Can approximate functions from dataset



Neuro Fuzzy

- Fuzzy logic does not require a mathematical model of the process, it just needs expert knowledge to formulate rules and membership functions.



IF temperature IS very cold THEN fan_speed is stopped
IF temperature IS cold THEN fan_speed is slow
IF temperature IS warm THEN fan_speed is moderate
IF temperature IS hot THEN fan_speed is high

IF-THEN rules map input or computed truth values to desired output truth values

- Neural networks are very good at acquiring “knowledge” from a dataset.
- Neural network can be connected to a fuzzy logic controller to create a very robust control system.

Current real world applications

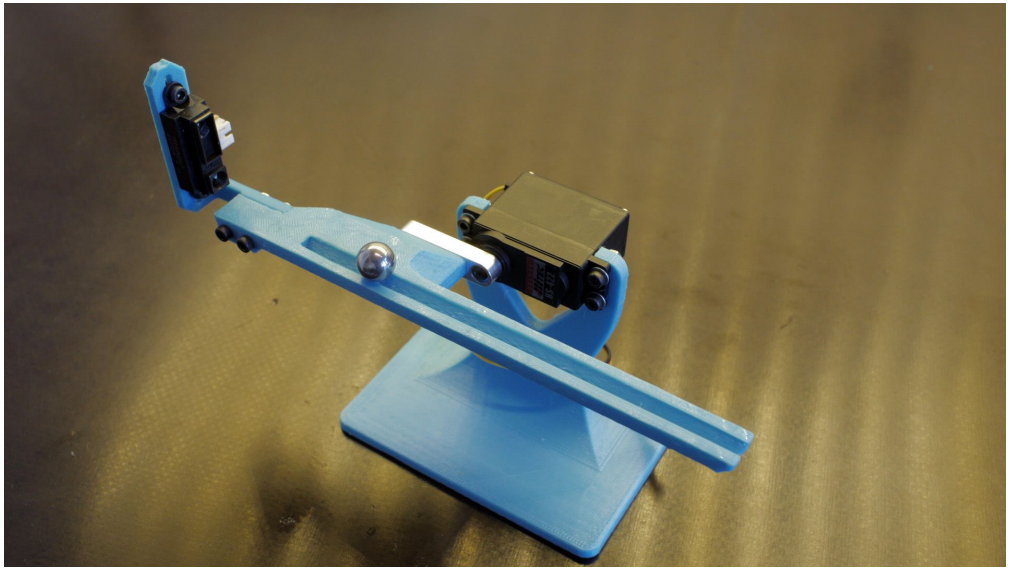
- Fuzzy logic controls the Sendai subway system in japan
    - Smart transmissions, braking systems, traffic planning, predicting customer usage of subway facilities.
- Efficient control of HVAC systems (air conditioning, thermostats, etc)
- Washing machines
    - Fuzzy logic controls water intake, water temperature, wash time, rinse performance, etc.

Fuzzy logic is adopted in Japan but not in Europe and America.

Test rig components

- Raspberry pi
- Servo motor
- IR distance sensor

TensorFlow



Aim is to keep ball a certain distance from IR sensor away by rotating the servo angle.

Timeline

- Research the subject and have good understanding by end of november
- Build test rig before christmas holidays with only Fuzzy Logic
- Add the neural network part after christmas holiday
- Experiment and compare with PID controller performance

References

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